

MONTICELLO, IA, USA

WMO: 725475

Lat: 42.224N

Lon: 91.166W

Elev: 849

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04953

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.5	(c) -3.2	(d) -16.5	(e) 2.3	(f) -8.3	(g) -10.5	(h) 3.2	(i) -2.5	(j) 28.3	(k) 18.1	(l) 25.5	(m) 19.0	(n) 4.4	(o) 290	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	90.5	76.6	88.1	75.4	85.7	73.9	79.6	86.8	77.6	85.4	75.7	83.1	10.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	146.4	84.5	75.0	135.5	82.8	73.0	126.3	80.7	43.9	86.8	41.5	85.4	39.8	83.1	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	20.7	18.5	-16.5	94.0	6.8	3.5	-21.4	96.5	-25.3	98.6	-29.1	100.5	-34.1	103.1		
			-17.8	82.4	7.5	1.9	-23.2	83.8	-27.5	84.9	-31.7	86.0	-37.1	87.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	20.7	24.0	37.2	48.5	60.6	70.0	72.8	70.8	63.5	50.8	37.8	25.6	(6)	(7)
	DBStd	20.80	12.48	12.89	11.65	9.67	8.51	6.21	5.56	5.34	8.08	9.36	10.97	11.78	(7)	(8)
	HDD50	3471	908	730	423	140	11	0	0	0	5	105	392	757	(8)	(9)
	HDD65	6726	1373	1149	863	501	193	23	5	11	121	448	818	1221	(9)	(10)
	CDD50	2986	0	1	26	94	339	601	708	646	411	132	26	2	(10)	(11)
	CDD65	765	0	0	1	4	56	174	249	192	77	9	3	0	(11)	(12)
	CDH74	7335	0	0	12	98	567	1640	2399	1691	767	100	61	0	(12)	(13)
	CDH80	2340	0	0	0	17	153	544	835	511	228	18	34	0	(13)	(14)
(14) Wind	WSAvg	7.6	8.7	8.7	9.1	9.8	8.7	7.0	4.9	4.7	5.6	7.5	8.2	8.1	(14)	(15)
(15) Precipitation	PrecAvg	38.0	1.3	1.4	2.2	3.7	4.7	5.6	4.9	4.4	3.3	2.8	2.2	1.6	(15)	(16)
	PrecMax	52.1	2.4	3.8	5.6	7.7	8.7	9.3	12.3	8.7	6.3	7.4	6.5	3.6	(16)	(17)
	PrecMin	27.1	0.3	0.1	0.4	1.2	0.7	1.2	0.6	0.4	0.7	0.5	0.2	0.2	(17)	(18)
	PrecStd	6.3	0.6	0.9	1.5	1.6	2.0	2.2	2.8	2.2	1.4	2.0	1.7	0.8	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	60.7	75.1	82.0	89.7	92.3	97.0	91.6	90.4	82.2	83.6	59.3	(19)	(20)
		MCWB	44.4	50.9	61.9	63.8	72.2	75.8	77.4	76.7	75.3	66.1	77.6	54.3	(20)	(21)
	2%	DB	45.0	50.5	67.6	77.0	84.0	89.9	90.5	88.3	86.2	76.8	65.9	50.2	(21)	(22)
		MCWB	40.4	45.2	57.2	61.2	68.2	74.8	78.6	76.5	73.0	62.3	56.7	45.5	(22)	(23)
	5%	DB	39.2	44.9	61.0	71.8	80.8	86.2	87.9	85.6	81.9	72.1	59.2	45.3	(23)	(24)
		MCWB	36.1	40.2	53.9	57.6	66.1	72.8	76.9	75.6	69.7	61.3	51.6	41.7	(24)	(25)
	10%	DB	36.3	39.3	54.5	65.8	76.6	82.3	84.3	82.1	78.8	67.6	54.3	40.5	(25)	(26)
		MCWB	34.0	35.8	47.8	54.9	64.1	70.5	74.8	72.8	66.9	59.0	48.4	37.1	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.2	52.2	64.4	67.4	74.8	80.2	83.0	81.0	77.9	71.9	74.7	56.2	(27)	(28)
		MCDB	48.5	59.0	72.6	79.3	84.9	88.1	90.5	88.7	86.7	79.8	83.5	60.5	(28)	(29)
	2%	WB	40.2	45.7	59.5	63.3	71.5	77.3	80.1	78.7	75.5	66.6	58.0	46.4	(29)	(30)
		MCDB	44.6	50.6	64.3	72.9	80.7	86.1	87.0	85.8	83.3	71.8	64.1	48.4	(30)	(31)
	5%	WB	36.7	40.2	54.0	59.7	68.6	75.1	78.1	76.4	72.6	63.0	53.3	41.4	(31)	(32)
		MCDB	39.7	44.3	60.8	69.4	76.6	83.0	85.5	83.2	78.8	69.0	58.5	44.8	(32)	(33)
	10%	WB	34.0	36.4	48.3	55.9	66.2	72.8	76.1	74.4	69.7	59.3	48.2	37.2	(33)	(34)
		MCDB	36.0	39.6	54.2	65.2	74.0	80.8	83.5	80.7	74.8	66.1	54.1	39.8	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	16.8	18.3	21.7	21.3	20.0	20.1	20.5	23.4	21.5	18.2	15.0	(35)	(36)
		MCDBR	19.0	20.8	26.7	30.6	25.0	23.3	22.0	22.7	26.0	27.5	26.0	18.8	(36)	(37)
	5% WB	MCWBR	15.4	15.9	17.4	17.5	13.1	12.2	12.7	14.0	14.3	15.8	18.4	15.4	(37)	(38)
		MCWBR	17.9	20.1	24.6	26.1	21.6	20.1	20.1	19.4	21.4	22.1	24.3	17.7	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.284	0.299	0.338	0.377	0.409	0.427	0.438	0.415	0.386	0.339	0.297	0.281	(39)	(40)
		taud	2.391	2.357	2.365	2.279	2.254	2.245	2.206	2.274	2.323	2.456	2.497	2.430	(40)	(41)
	Ebn at Noon		274	287	288	283	275	269	265	268	267	270	268	266	(41)	(42)
		Edh at Noon	25	30	35	41	43	43	44	40	36	28	23	22	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		568	885	1197	1477	1748	1939	1944	1706	1426	929	619	453	(43)	(44)
	RadStd		63	89	117	121	123	167	127	113	113	113	48	43	(44)	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page